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BENCHMARKING OF THE SYMMETRIC KEY ALGORITHMS AS AES USING THE SAME ARCHITECTURE

OBJECTIVE

- Implement and benchmark AES encryption/decryption using the Nexys A7 (Artix-7 FPGA)
- Evaluate performance through using a single, consistent hardware architecture
- Measure and Compare:

- Throughput
- Latency
- Resource Utilization
- Power consumption

- FPGA Offers:

- Parallel hardware implementation
- Customizable data path width
- Accurate Hardware-level benchmarking
- No OS or background process interface

- About the Board:

-FPGA: Xilinix Artix-7

-Clock Source: 100 MHz Oscillator

-Development tool: Vivado

-Interface: UART/ USB/ switches for input-output testing

IMPLEMENTATION APPROACH

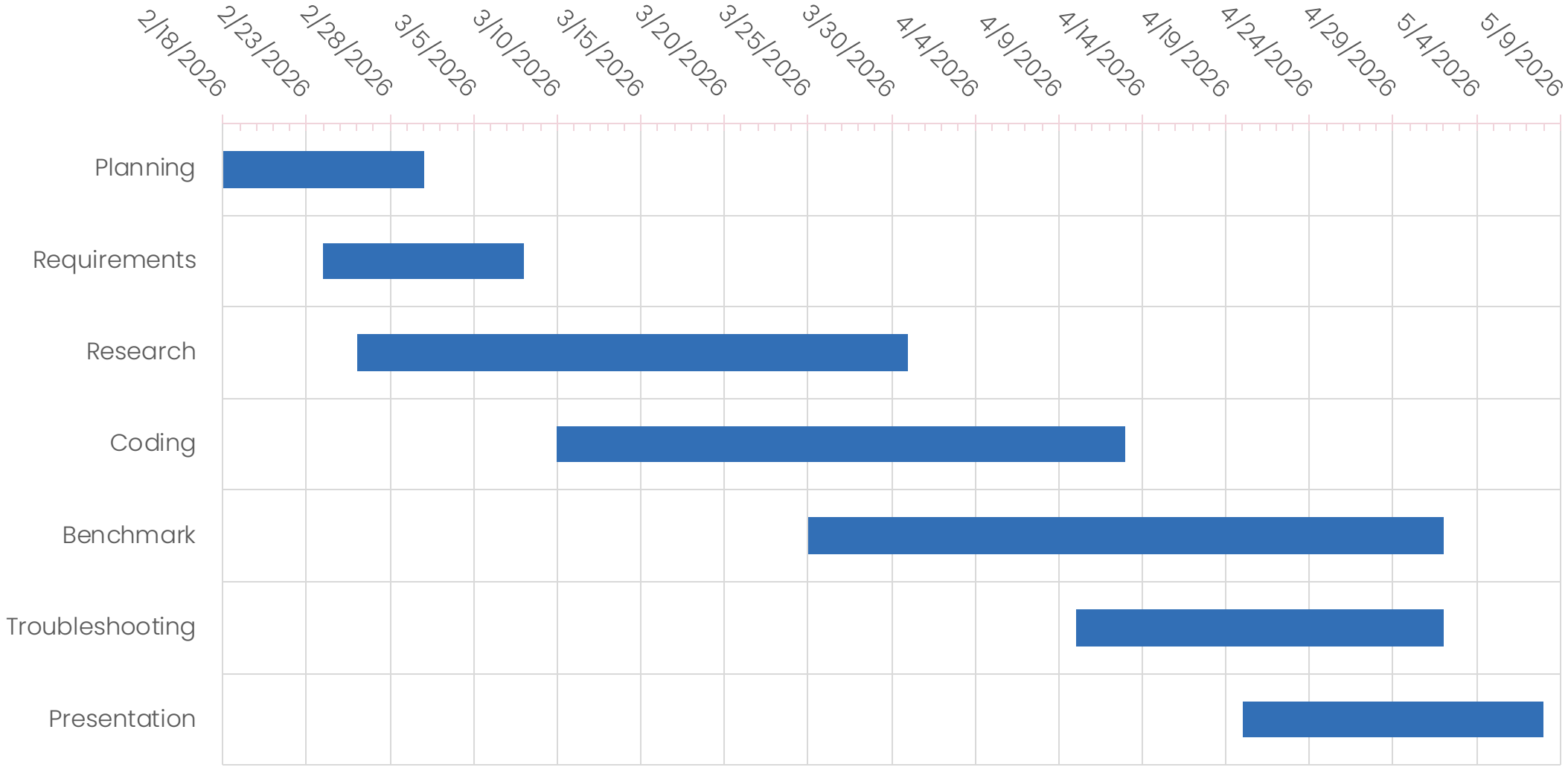
- Develop AES-128 in Verilog/VHDL
- Use identical architecture for all cases:

- Same clock frequency
- Same I/O interface
- Same data test size

- Performance Metrics:

- Throughput: $Throughput = \frac{Block\ size\ x\ Clock\ Frequency}{Clock\ cycles\ per\ block}$
- Latency: Number of clock cycles required to encrypt one 128-bit block
- Resource Utilization: LUTs, Flip Flops, BRAM, DSP slice(all will come from VIvado)
- Power consumption: Done using Vivado Power Analyzer

GANTT CHART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Planning
Requirements
Research
Coding
Benchmark
Troubleshooting
Presentation